



Healthcare
Analytics
project:

Healthcare Performance Analytics Dashboard

Total Doctors

1K



Total Patients

10K



Total Revenue

25.41M



Total Visits

10K



Avg Visit Cost

2.54K



Year

- ☐ 2023
☐ 2024
☐ 2025

Total Visits by Month

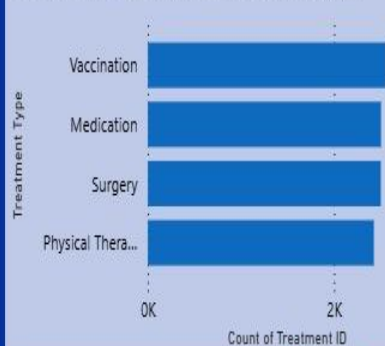


State and Treatment Cost

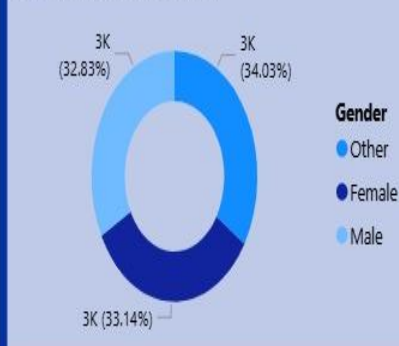
State

- ☐ California
☐ Florida
☐ Illinois
☐ New York
☐ Texas

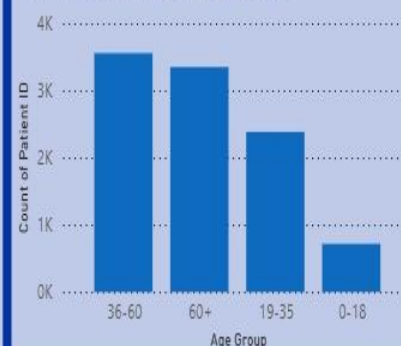
Count of Treatment ID by Treatment Type



Total Patients by Gender



Count of Patient ID by Age Group



Power BI: Healthcare Analytics

- ◆ This Healthcare Performance Analytics Dashboard provides a comprehensive overview of hospital performance and operational efficiency. It highlights key performance indicators such as total doctors (1K), total patients (10K), total revenue (25.41M), total visits (10K), and the average visit cost (2.54K), giving a quick snapshot of overall performance. The monthly visits trend shows fluctuations throughout the year, with peak visits observed around August and October, indicating seasonal demand patterns. The gender distribution chart shows a nearly balanced ratio among male, female, and other patients. Age group analysis indicates that the majority of patients fall within the 36–60 age group, followed by senior citizens, while the 0–18 group has the lowest count. Additionally, year and state filters allow dynamic analysis across different time periods and locations, making the dashboard interactive and useful for data-driven decision-making.

MySQL:

```
1 • CREATE DATABASE healthcare_db;
2 • USE healthcare_db;
3
4 • CREATE TABLE patient (
5     PatientID INT,
6     Gender VARCHAR(10),
7     DateOfBirth DATE,
8     Age INT,
9     PhoneNumber VARCHAR(20),
10    Address TEXT,
11    BloodType VARCHAR(5),
12    EmergencyContact VARCHAR(100),
13    InsuranceProvider VARCHAR(100),
14    State VARCHAR(50),
15    City VARCHAR(50),
16    Country VARCHAR(50),
17    PolicyNumber VARCHAR(50),
18    MedicalHistory TEXT
19 );
20
```

- ❖ The **Healthcare Database Project** is designed to manage and analyze hospital data using **SQL**. This database stores important information about **patients, doctors, visits, and treatments**.
- ❖ Using SQL queries, we can easily retrieve and analyze data to understand **patient records, doctor activities, treatment costs, and disease patterns**. This helps healthcare organizations manage their data more efficiently.

```
22 • select * from patient;
23
24 • SELECT COUNT(*) AS total_patients
25 FROM Patient;
26
27 • SELECT COUNT(*) AS total_visits
28 FROM Visit;
29
30 • SELECT SUM(cost) AS total_revenue
31 FROM Treatment;
32
33 • ALTER TABLE Visit
34 CHANGE `i»¿Visit ID` visit_id INT;
35
36 • ALTER TABLE doctor
37 CHANGE `i»¿Doctor ID` doctor_id INT;
38
```

The main objectives of this project are:

- ❖ To create a **structured healthcare database**.
- ❖ To store and manage patient and hospital information efficiently.
- ❖ To analyze healthcare data using **SQL queries**.
- ❖ To generate insights such as **total patients, doctor visits, treatment costs, and common diagnoses**.

```

39 • ALTER TABLE Visit
40 CHANGE `Doctor ID` doctor_id INT;
41
42 • SELECT d.`Doctor Name`, COUNT(v.visit_id) AS total_visits
43 FROM Visit v
44 JOIN Doctor d ON v.doctor_id = d.doctor_id
45 GROUP BY d.`Doctor Name`;
46
47 • select * from doctor;
48 • select * from visit;
49 • SELECT city, COUNT(*) AS patient_count
50 FROM Patient
51 GROUP BY city;
52
53 • SELECT Diagnosis, COUNT(*) AS Frequency
54 FROM Visit
55 GROUP BY Diagnosis
56 ORDER BY Frequency DESC

```

```

57 LIMIT 5;
58
59 • SELECT AVG(Cost) AS Avg_Treatment_Cost
60 FROM Treatment;
61
62 • select * from visit;
63 • select * from doctor;
64

```

- ◈ this healthcare SQL project demonstrates how databases can be used to manage and analyze hospital data effectively.
- ◈ By using SQL queries, we were able to identify total patients, doctor workloads, treatment revenue, patient distribution, and common diseases.
- ◈ These insights help healthcare organizations improve decision-making, resource allocation, and patient care management.

Excel:

Patient Healthcare Dashboard

Total Patient
10K

Total Doctors
1000

Total visit
10K

Average Age of Patient
48

Avg. Treatment cost
\$524.75

Top 10 Patient visit in Doctor

Top 5 Diagnost patient

Treatment Name And ID

dosage of
medician

Test Result by Ref. Range

City wise condition

Treatment Name

Antibiotic

Ibuprofen

Metformin

Omeprazole

Physical Therapy

Dosage

10mg

20mg

50mg

5mg

City

Chicago

Houston

Los Angeles

Miami

- ❖ This **Patient Healthcare Dashboard** provides a visual overview of hospital data related to patients, doctors, treatments, and medical conditions. It helps healthcare management quickly understand important insights and make better decisions.
- ❖ the **Patient Healthcare Dashboard** helps visualize and analyze hospital data effectively. It provides insights about patient visits, doctor workload, treatment costs, diagnoses, and regional health trends.
- ❖ By using this dashboard, healthcare organizations can make **data-driven decisions, improve resource management, and enhance patient care services.**

Tableau:

- ◆ This Tableau dashboard analyzes **healthcare data** to understand patient visits, revenue, and treatment patterns.
- ◆ This healthcare dashboard provides a complete overview of hospital performance by analyzing patient visits, revenue, diagnosis patterns, and demographics. It helps healthcare management identify trends, improve resource allocation, and make data-driven decisions to enhance patient care and operational efficiency.

Challenges Face:

- ◆ The first challenge was cleaning the data because some values were missing or not properly formatted.
- ◆ Another challenge was combining different data like patients, doctors, visits, and treatments.
- ◆ Writing accurate SQL queries for analysis also required careful checking.
- ◆ While building dashboards, choosing clear visuals was important.
- ◆ Finally, I focused on making the dashboard simple and interactive for users.